

Solution Architecture

Date	30 September 2026	Team ID	6ab3e57da238dc999eb3d44d
Project Name	EduGenie – AI-Powered Learning Assistant	Maximum Marks	As per template

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GitHub: <https://github.com/ricardovictor-28507/Google-Cloud-Generative-AI--A-S>

Architecture Flow

Layer	Components	Responsibility
Presentation	templates/index.html, static/app.js, static/style.css	Collect input, select tool, call API, render results.
API	main.py / FastAPI	Routes requests, validates data, handles errors, exposes OpenAPI.
Feature Logic	qna.py, explanation_module.py, summary_module.py, quiz_module.py, learning_path.py	Build task-specific prompts and process feature results.
AI Integration	ai_service.py	Configure Gemini client, send prompts, provide fallback.
External Service	Google Gemini API	Generate educational content when a live key/model is available.

Request flow: Browser → FastAPI → Feature module → AIService → Gemini/demo fallback → Feature result → Browser.